



University of AJ&K, Neelum Campus

Department of Computer Science & IT

Computer Networks

Assignment No 2

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Submitted To:

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Submission Date: 4-Sept-2025

Networking Commands

1. `ipconfig`

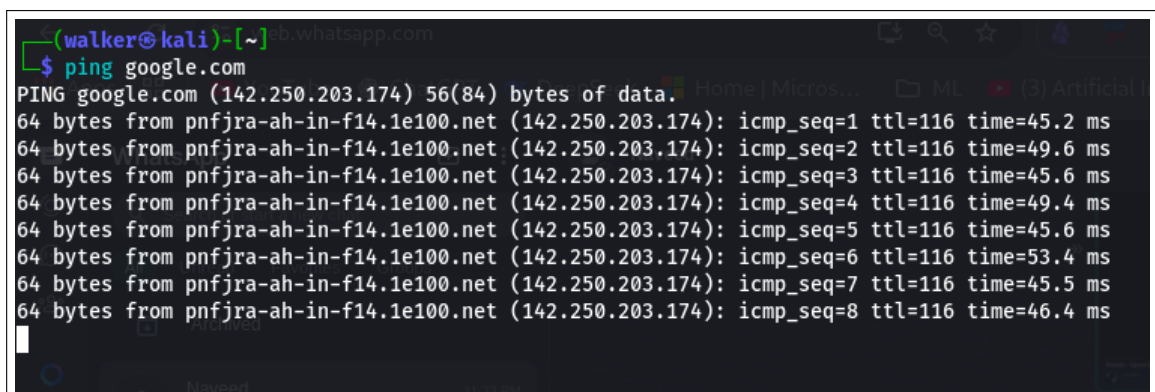
- **Purpose:** To display the basic IP configuration for all network adapters on a Windows machine.
- **Observation:** Shows my IP address, subnet mask, and default gateway for each adapter.

2. `ipconfig /all`

- **Purpose:** To display the full, detailed TCP/IP configuration for all adapters.
- **Observation:** Shows all basic info plus my MAC address, DHCP server, DNS servers, and lease information.

3. `ping <IP>`

- **Purpose:** To test connectivity to another network device and measure response time.
- **Observation:** Sends packets from my machine to the target IP and displays whether replies are received and how long they take.



```
(walker@kali)-[~] b.whatsapp.com
$ ping google.com
PING google.com (142.250.203.174) 56(84) bytes of data:
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=1 ttl=116 time=45.2 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=2 ttl=116 time=49.6 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=3 ttl=116 time=45.6 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=4 ttl=116 time=49.4 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=5 ttl=116 time=45.6 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=6 ttl=116 time=53.4 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=7 ttl=116 time=45.5 ms
64 bytes from pnfjra-ah-in-f14.1e100.net (142.250.203.174): icmp_seq=8 ttl=116 time=46.4 ms
```

4. `arp -a`

- **Purpose:** To display the ARP table, which maps IP addresses to MAC addresses on the local network.

- **Observation:** Lists the IP and MAC addresses of devices my computer has recently communicated with.

5. **tracert** < *hostname* >

- **Purpose:** To trace the path packets take from my computer to a destination host.
- **Observation:** Lists all routers (hops) between my computer and the destination, showing the latency at each step.

6. **nslookup**

- **Purpose:** To query DNS servers to find the IP address of a domain name or the domain name of an IP address.
- **Observation:** Shows the DNS server my computer is using and the results of the query.

7. **netstat -a**

- **Purpose:** To display all active network connections and listening ports on the computer.
- **Observation:** Shows a list of all TCP and UDP connections to and from my computer and their current state.

8. **netstat -r**

- **Purpose:** To display the kernel's IP routing table.
- **Observation:** Shows the routing table on my computer, including network destinations, gateways, and interfaces.

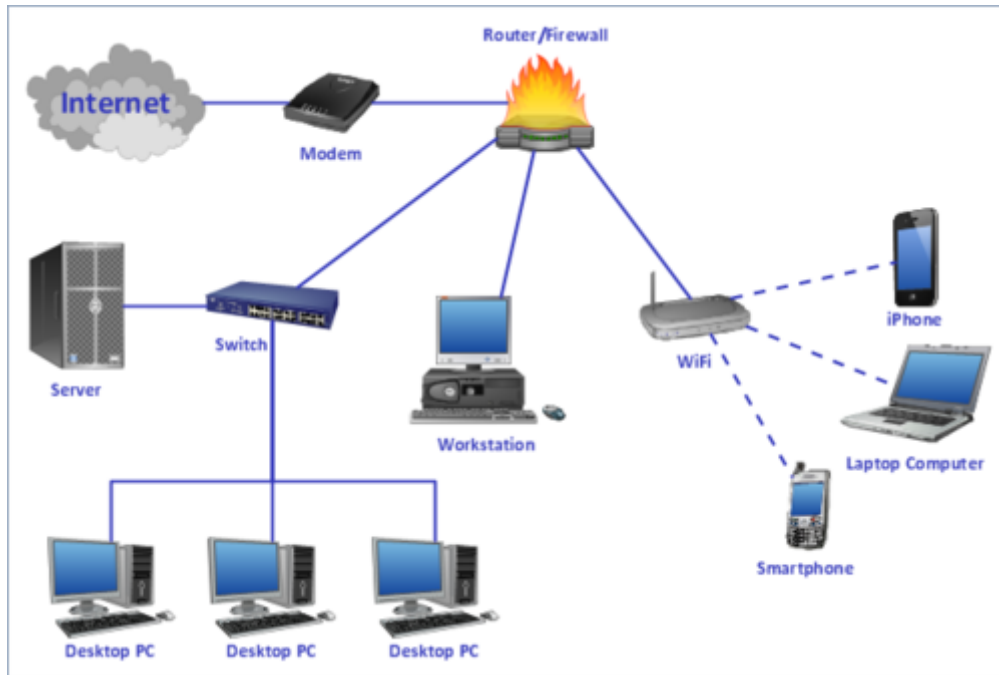
9. **net view**

- **Purpose:** To display a list of computers and shared resources available on the network.
- **Observation:** Lists the names of other computers and servers visible on my local network.

10. **nbtstat -n**

- **Purpose:** To display NetBIOS names registered by the local computer.
- **Observation:** Shows the NetBIOS name table for my computer, including the registered names and services.

Networking Devices



1. Find out the end devices and intermediary devices depicted in the scenario and place them rightly.

End Devices	Intermediary Devices
PC	Switch
Laptop	Router
Server	Wireless Router / Access Point
Printer	Cloud (representing WAN)
IP Phone	

2. What is the difference between a hub, switch and a router? (Be specific).

Device	Function & Difference
Hub	Operates at Layer 1 (Physical). Broadcasts all data to every port, creating congestion.
Switch	Operates at Layer 2 (Data Link). Learns MAC addresses and sends data only to the specific destination port, reducing traffic.
Router	Operates at Layer 3 (Network). Connects different networks together and uses IP addresses to route packets between them.

3. Which networking devices are used in layer 1, 2 and 3 of TCP/IP model?

Layer #	Layer Name	Devices used
1	Network Interface	Hub, Repeater, Cables, Modem
2	Internet	Switch, Bridge, Network Card (NIC)
3	Transport/Application	Router, Multilayer Switch

4. My company wants to purchase some network hardware to which they can plug the 30 PCs in my department. Which type of network device is appropriate?

- a) A router
- b) A firewall
- c) A switch
- d) A server

Correct Option: c) A switch

Explanation: A switch is designed to connect multiple devices within a single Local Area Network (LAN) and efficiently manage traffic between them.

5. I received a video file from my friend's Apple iPhone using AirDrop. What was his iPhone functioning as in that transaction?

- a) A server
- b) A client
- c) A Local Area Network

Correct Option: a) A server

Explanation: The iPhone was hosting and sending (serving) the file to my device, making it the server in this client-server transaction.

6. What is my computer or smart phone functioning as when I watch a video from YouTube?

- a) A server
- b) An end-host

c) A client

Correct Option: c) A client

Explanation: My device is requesting and receiving data from YouTube's servers, which makes it a client.

7. My company wants to purchase some network hardware to connect its separate networks together. What kind of network device is appropriate?

a) A firewall

b) A host

c) A LAN

d) A router

Correct Option: d) A router

Explanation: A router's primary function is to connect different networks and route traffic between them.